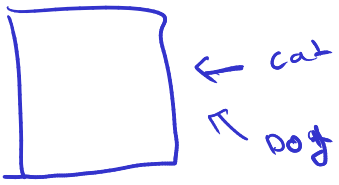
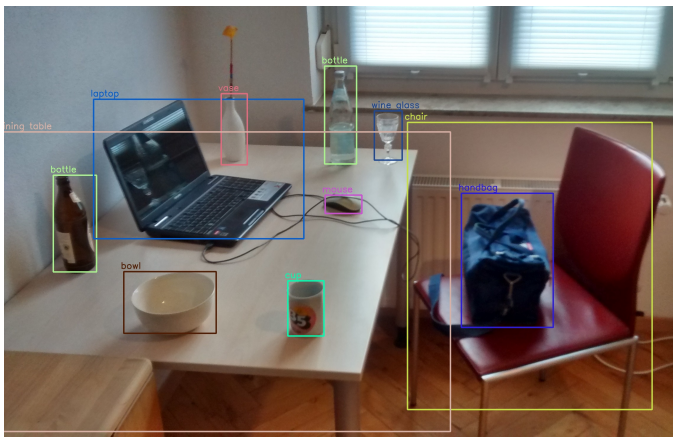
Object Detection

CNN → classify



CNN → Image classification

Object Detection



↳ class - classify
↳ position of the object

Bounding Box

Object Detection

- 2 stage Detector
  - → RCNN → Regional based CNN
  - → Faster RCNN
  - etc.
- 1 stage Detector
  - SSD - Single Stage Detector
  - YOLO - You look only once

1. Transformer Object Detection
2. Agentic object Detection

Why?

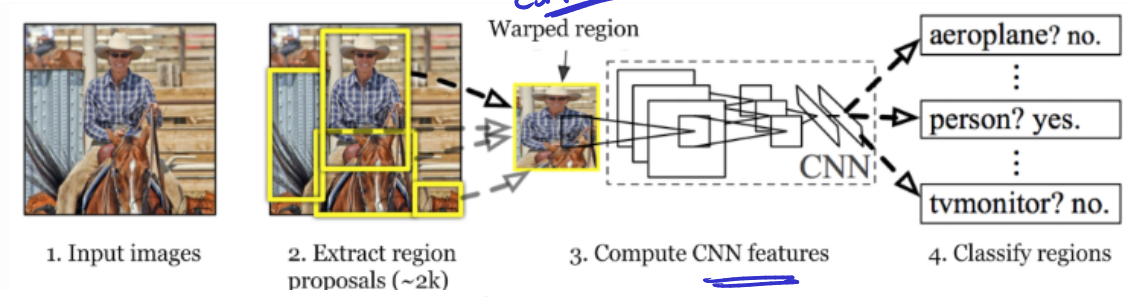
1. Document understanding
2. Count
3. Self driving cars

2 stage Detector

Regional based CNN

Selective search

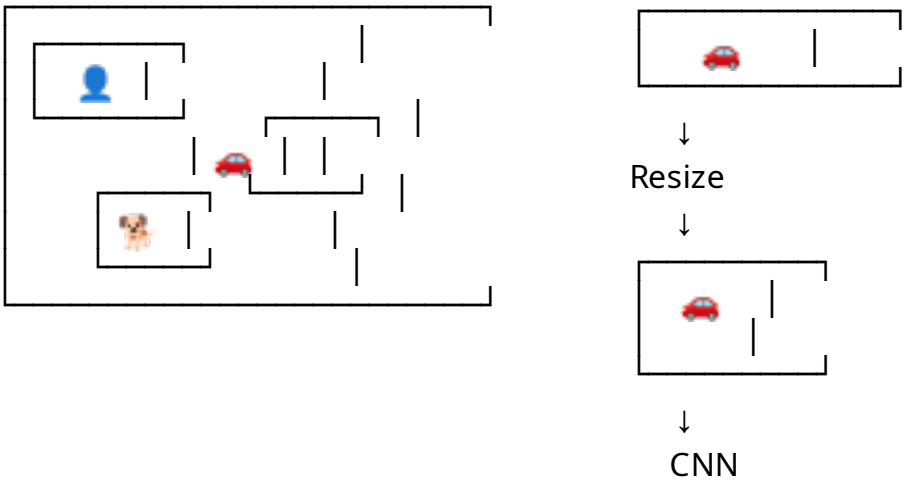
cvpr site



Probability

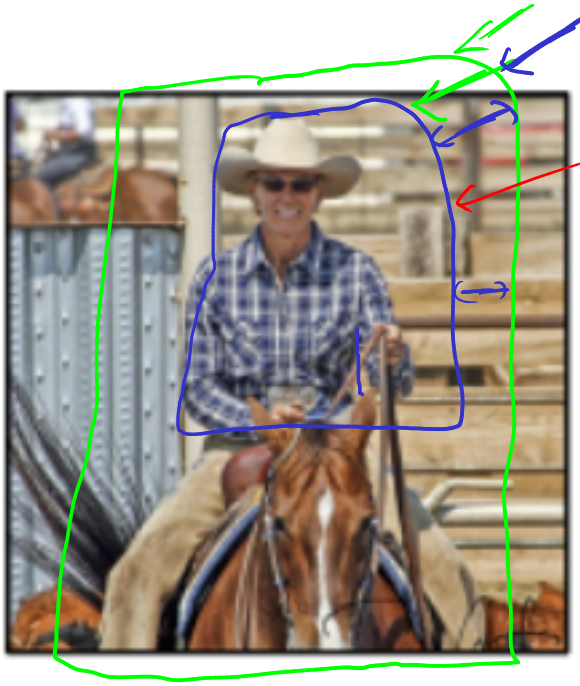
Threshold

2000



Bounding Box Regression

Numerical Data → Regression



Object Detection

annotate the bounding boxes

Annotation

Roboflow → tool for annotation

Regression

error → MSE → Mean square error

$$\sum_{i=1}^{n} \frac{(y_i - y_i')^2}{n}$$

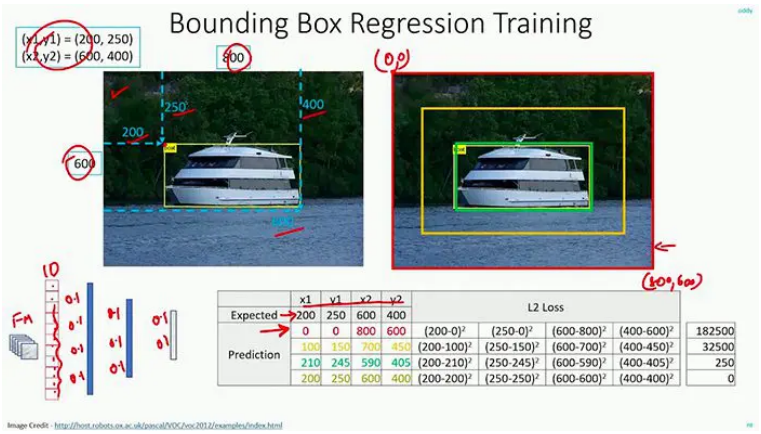
y → Actual value

y' → predicted value

Actual BB x_1, y_1, x_2, y_2

10, 20, 50, 100 → error

Region BB 5, 10, 80, 200



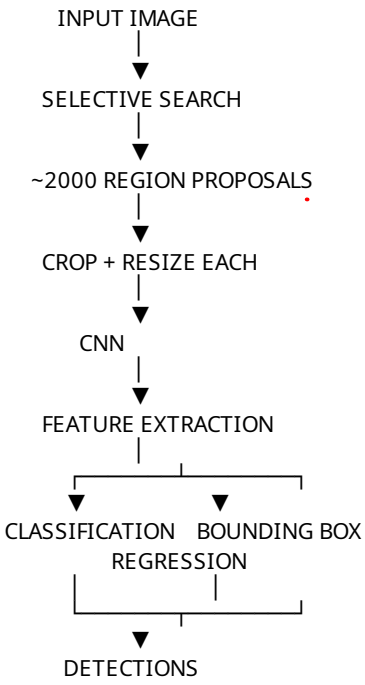
	x1	y1	x2	y2	L2 Loss				
Expected →	200	250	600	400					
Prediction →	0	0	800	600	$(200-0)^2$	$(250-0)^2$	$(600-800)^2$	$(400-600)^2$	182500
	100	150	700	450	$(200-100)^2$	$(250-150)^2$	$(600-700)^2$	$(400-450)^2$	32500
	210	245	590	405	$(200-210)^2$	$(250-245)^2$	$(600-590)^2$	$(400-405)^2$	250
	200	250	600	400	$(200-200)^2$	$(250-250)^2$	$(600-600)^2$	$(400-400)^2$	0

Image Credit - http://host.robots.ox.ac.uk/pascal/VOC/voc2012/examples/index.html

2 Stage → 1 stage → CNN

2stage → Bounding Box Regression

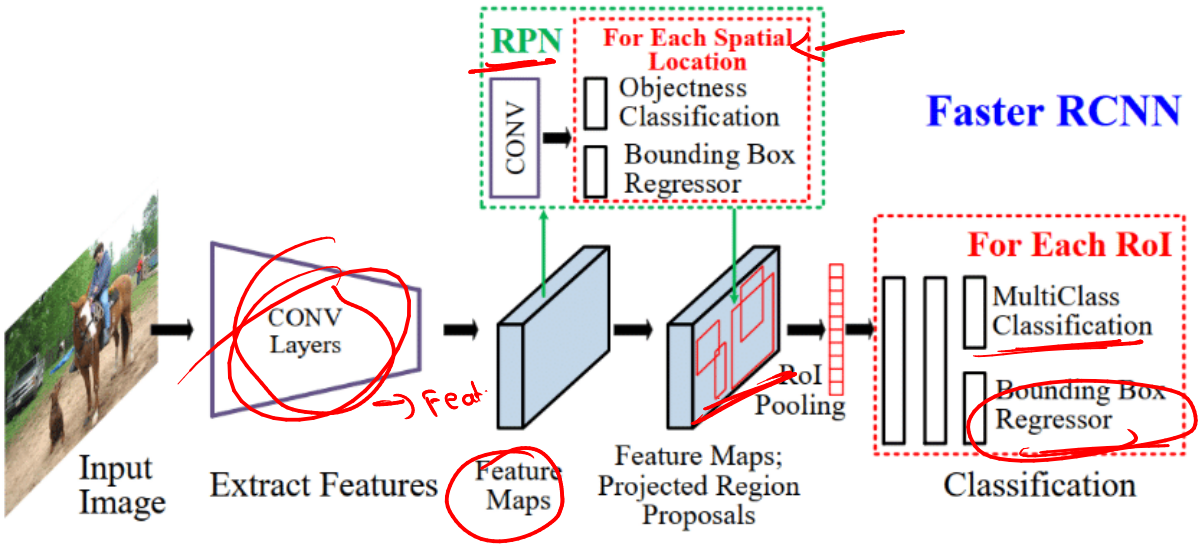
Accrate but slow



Huge computation

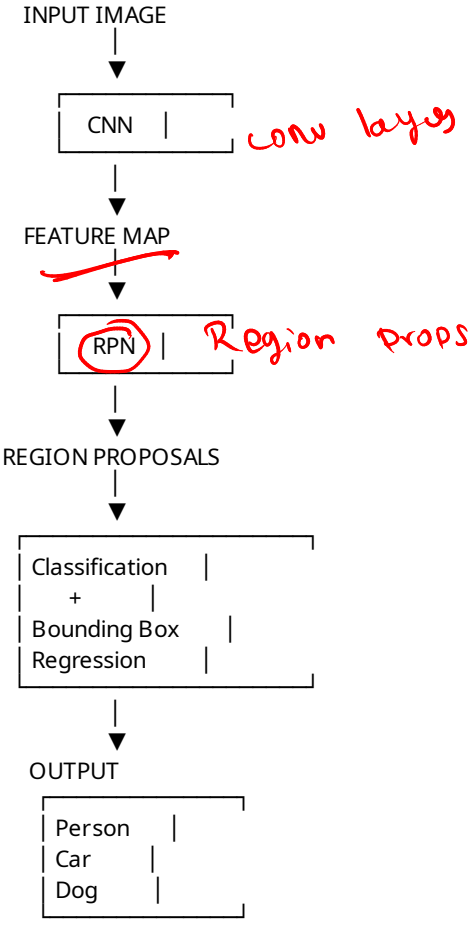
Faster RCNN

Region proposal Network



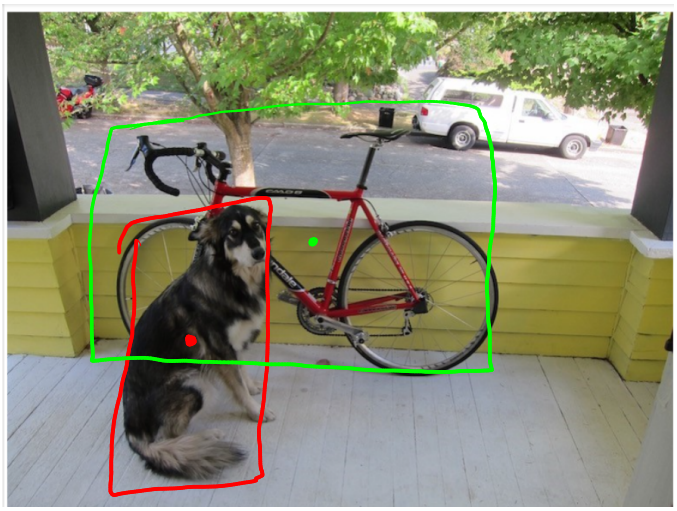
RPN → is a specilized NN → Trained to identify Region from Feature maps

2stage → Extracting Region
Classification
↳ Regression



1stage Detector → Classification & Bounding Box Detection will happen at once

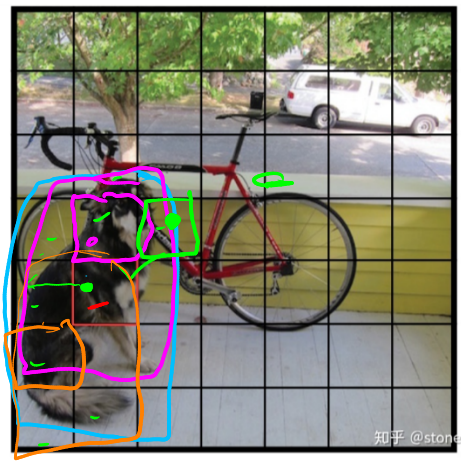
YOLO → You look only once



Grid ⇒

7×7

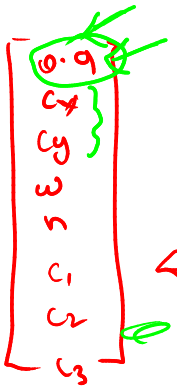
49 cell



each grid is send to Algorithm

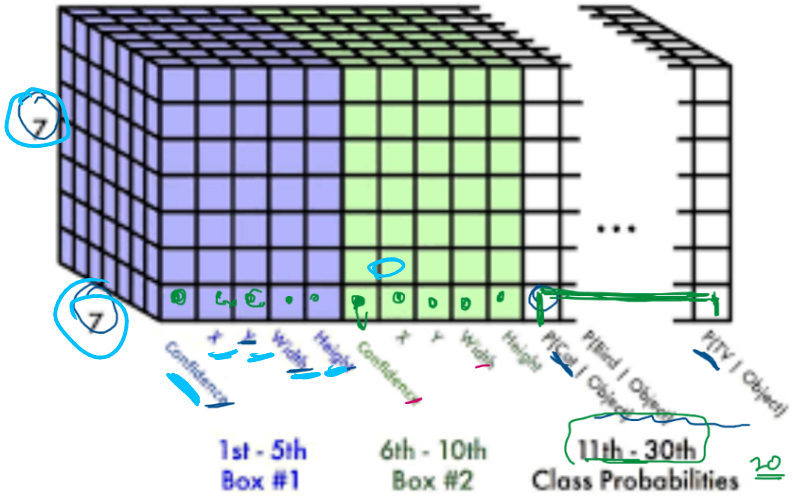
1. Weather ther is an object in Image
2. Wheather it has center of the object
3. Which class it belongs to

Grid cell → Objectness score
Bounding box
Class probability

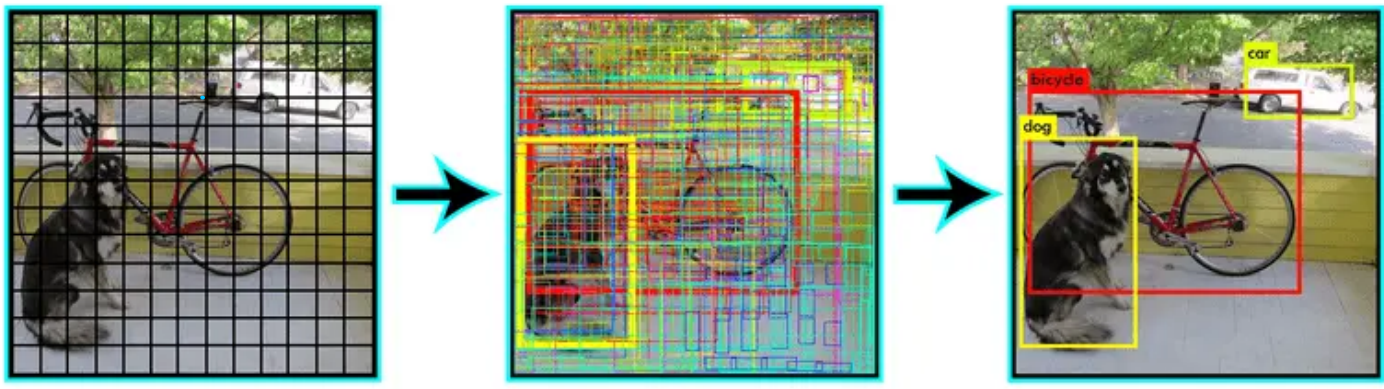


Yolo return for each grid cell

Objectness score → what is the probability that it has center of the object



L = object loss + BB loss + class loss error



Filtering & NMS

1. Remove all the boxes, with low objectness score

confidence threshold 0.5

Boxes < 0.5

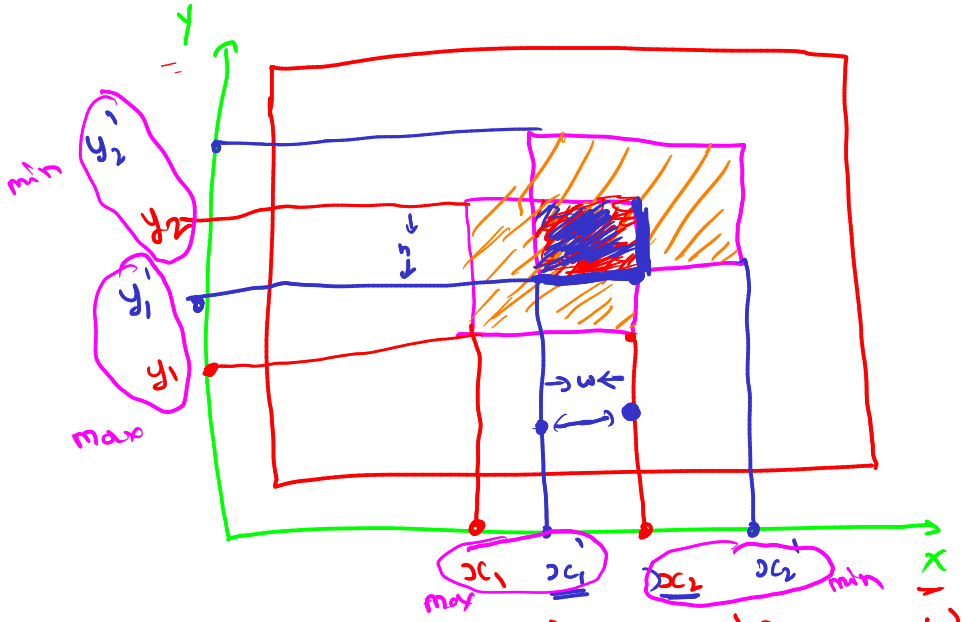
2. NMS → Non Max suppression

IOU → Intersection over union

IOU → Intersection over union

↳ metric to identify the Box quality
↳ overlapping boxes

IOU → Ratio of Intersect w.r.t union



Area of overlap
IOU = Area of union

IOU → 0 to 1

$w = x_2 - x_1'$

$h = y_2 - y_1'$

Area of rectangle = w * h

Area of overlap = $w_o * h_o$

Bounding Box = x_1, y_1, x_2, y_2

Bounding Box2 = x_1', y_1', x_2', y_2'

$I_{x_1} = \max(x_1, x_1')$

$I_{x_2} = \min(x_2, x_2')$

$I_w = I_{x_2} - I_{x_1}$
↳ width of Intersec

$I_h = I_{y_2} - I_{y_1}$
↳ height of Intersec

$I_{y_1} = \max(y_1, y_1')$

$I_{y_2} = \min(y_2, y_2')$

Area of overlap ⇒ $I_w * I_h$

Area of Union
↳ Area of Box1 + Area of Box2 − (Area of overlap)

Area of Box-1

x_1, y_1, x_2, y_2

$w = x_2 - x_1$

$h = y_2 - y_1$

Area

$Box1 = w \times h$

Area of Box 2

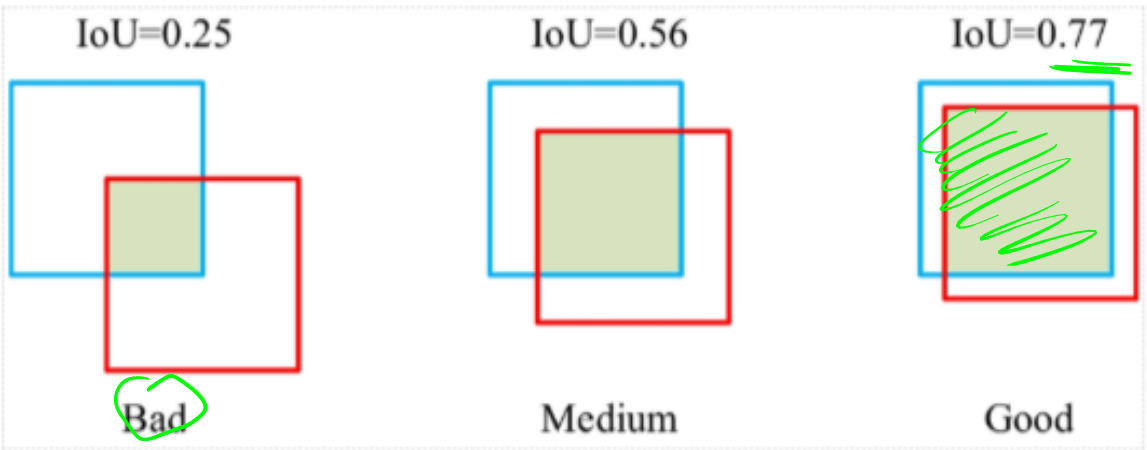
x_1', y_1', x_2', y_2'

$w' = x_2' - x_1'$

$h' = y_2' - y_1'$

$Box2 = w' \times h'$

$$IoU = \frac{\text{Area of overlap}}{\text{Area of union}}$$



NMS

↳ Non Max suppression

Remove overlapping boxes of same class

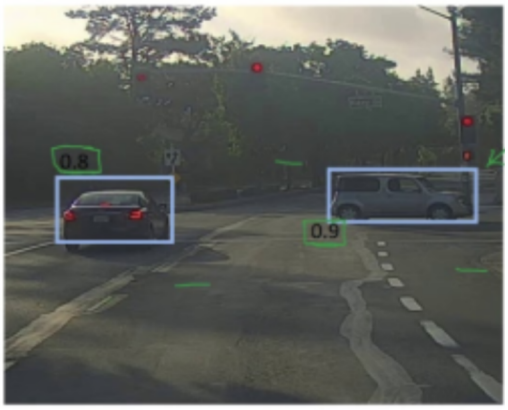


For every class

1. Box with highest confidence → slected Box
2. compute IOU with selected Box with other Box
3. if IOU > 0.4 ⇒ overlapping Boxes
4. Remove the overlapping Boxes
5. Repeat 1 to 4 for other → until all are over

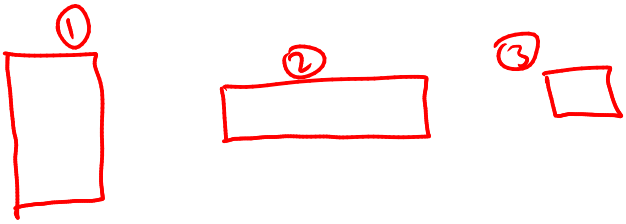
add 0.9 box to selected List

add 0.8 box to selected List



Anchor Boxes

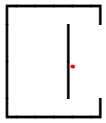
↳ Predefined Boxes shape



⇒ starting templat or reference shape

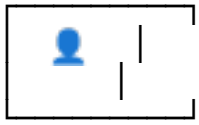
Anchor

↓



↓ regression

Final box



How Anchor Boxes are identified

Training Data

Kmeans technique

3 - top clusters

avg → Anchor Boxes

Ultralytics

pip install ultralytics

boxes, confidence, class score

Ultralytics

PAR Inventory Management

→ Inventory management

Slot if the items are less / going to finish
↳ order

Yolo Training process

1. Collect the dataset
↳ atleast 1000 Images
↳ atleast 500 - 600 Images for each class

2. Annotate the Dataset
↳ Roboflow
→ Creat a project
→ upload the dataset to Roboflow
→ Annotate each Image by providing Bounding Box & class
→ Create a new version
↳ Provide the config parameters
→ Download the Dataset

3. Prepare the train script
4. Start the training

Transfer learning → start with pre-defined weights

$$mAP = \frac{1}{N} \sum_{i=1}^{n} AP_i$$

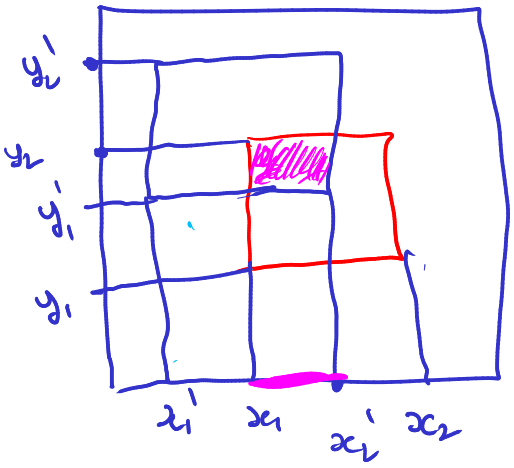
N - no of classes

AP - Average precision

$$Precision = \frac{TP}{TP + FP}$$

AP - 25.1.
AP = Product
AP - empty
AP = Full

$$mAP = \frac{1}{6} [AP_{25} + AP_{50} + AP_{Full} + \dots AP_{empty}]$$



$x_2' - x_1$